



Semiconductor relays provide the same function as electromechanical relays but have no moving parts, which increases their long-term reliability. These relays provide an array of solutions, meeting the needs of

Presented here is the design of such a relay for automotive applications that uses two series connected MOSFETs to achieve bidirectional conduction with advantages like:

1. No mechanical moving parts
2. No arching in contacts
3. No acoustical noise
4. No EMI from contacts commutation
5. High switching speed
6. High reliability

7. Long operating life
8. Resistance to shock and vibration

Fig. 1 shows the circuit of semiconductor relay for automotive applications. The circuit is built around switch-

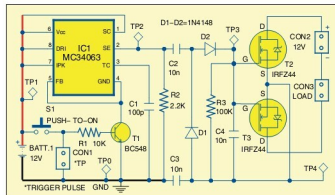


Fig. 1: Circuit of semiconductor relay for automotive applications

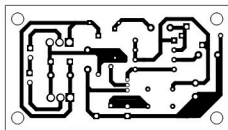


Fig. 2: An actual-size, single-side PCB for the semiconductor relay for automotive applications

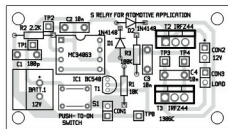


Fig. 3: Component layout for the PCB

BATT.1 - 12V battery
CON1-CON3 - Two-pin connector
S1 - Push-to-on switch

Test point	Details
TP0	0V, GND
TP1	12V
TP2	Oscillations (147 kHz approx.)
TP3	8.5V (approx.)

C4 or R3 at the expense of increased ripple on the rectified gate voltage.

When the MOSFETs are off, their parasitic diodes connect in series opposition and thus block conduction. You can select from a range of MOSFETs to match your application's requirements, but make sure that the voltage you apply to the gates of T2 and T3 is sufficient to fully switch both the devices into full conduction.

Construction and testing

An actual-size, single-side PCB for the semiconductor relay for automotive applications is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable case.

To test the circuit for proper functioning, verify 12V DC at test point TP1 with respect to TP0. Now press switch S1 and verify oscillations at TP2. Also check biasing voltage at TP3 with respect to TP4 as mentioned in the table. Please note that switch S1 is provided for testing purpose only. CON1 should be used for external trigger. ●

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